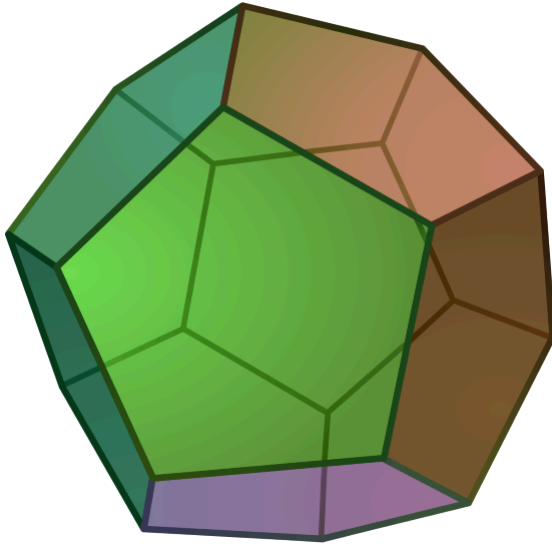


Question 23 – Stage 2

The 12 letters $D, O, D, E, C, A, H, E, D, R, O, N$ are randomly assigned to the 12 faces of a regular dodecahedron, one letter per face.

What is the probability (to 3.d.p.) that no two faces labelled D share an edge but the two faces labelled O do share an edge?

Working



[Image Source](#)

```
In [13]: from itertools import permutations
from math import factorial

neighbors = { # Using the numbers on the sides of the d12 above as reference
    1: {2,3,4,5,6},
    2: {1,3,7,8,4},
    3: {1,2,5,9,7},
    4: {1,2,8,6,10},
    5: {1,3,6,9,11},
    6: {1,4,5,10,11},
    7: {2,3,9,8,12},
    8: {2,4,7,10,12},
    9: {3,5,7,11,12},
    10: {4,8,6,11,12},
    11: {5,6,9,10,12},
    12: {9,8,7,10,11},
}

numberOfAlternates = factorial(12-5) # The number of possible arrangements of the other Letters for each arrangement of the Ds and Os
possiblePositions = range(1,13)
countTotal = 0
countOnlyOShares = 0

for Ds in permutations(possiblePositions,r=3):
    for Os in permutations(possiblePositions,r=2):
        if not any(o in Ds for o in Os): # Crude check to make sure every position is unique
            countTotal += numberOfAlternates

            # Check for neighbors
            neighboringDs = any(neighbors[d].intersection(Ds) for d in Ds)
            neighboringOs = any(neighbors[o].intersection(Os) for o in Os)

            if neighboringOs and not neighboringDs:
                countOnlyOShares += numberOfAlternates

print(f"Probability: {round(countOnlyOShares/countTotal,3)}")
```

Probability: 0.038